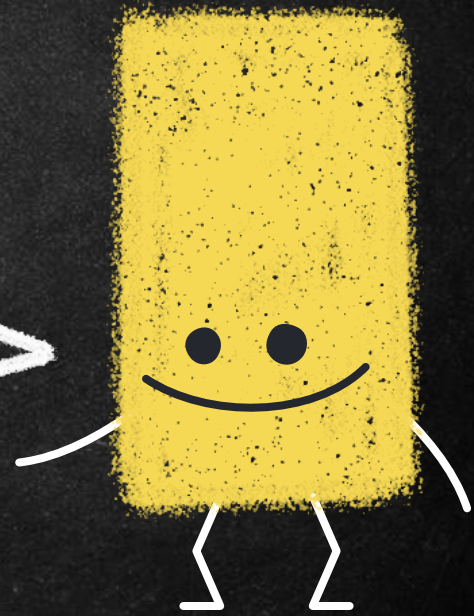
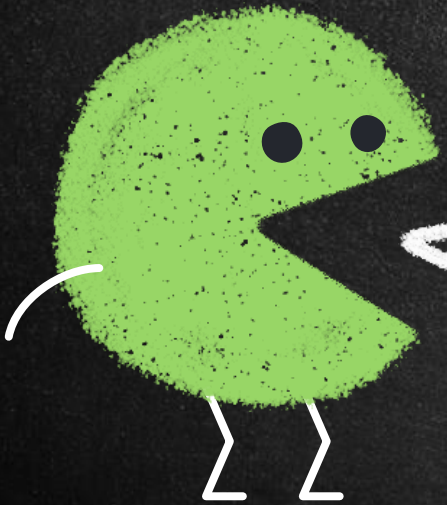


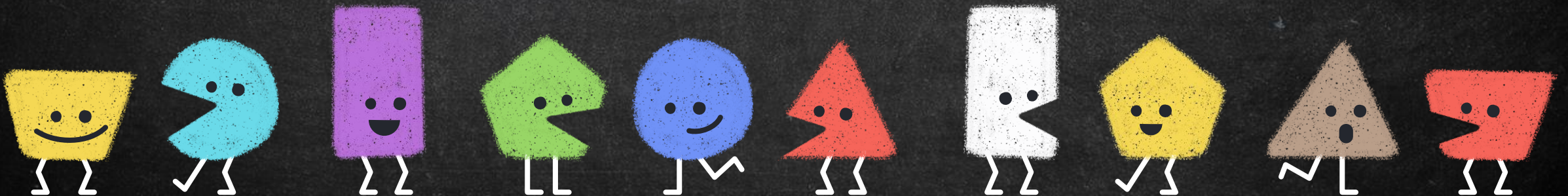
Digital Forensics Part - 3



Hello!

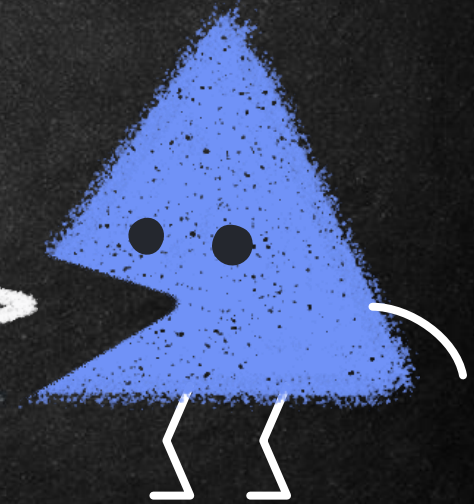
I am Sarang Rajvansh
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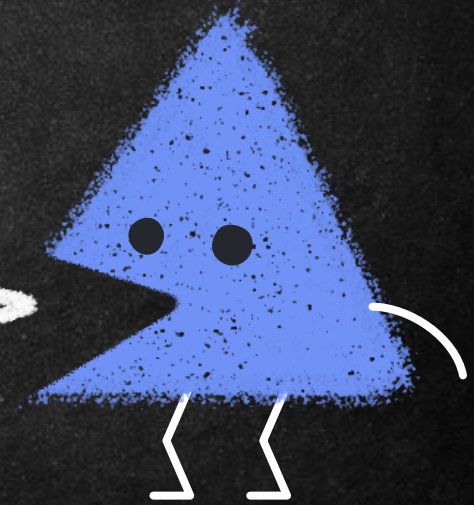
How Data is stored?

- Magnetic disk drives historically stored data in units known as cylinders, heads, and sectors. These units were then used to assign address to each storage block on a disk.
- concentric circles of storage areas on each side of the drive's platter were known as *tracks*. Tracks were limited to 63 sectors.



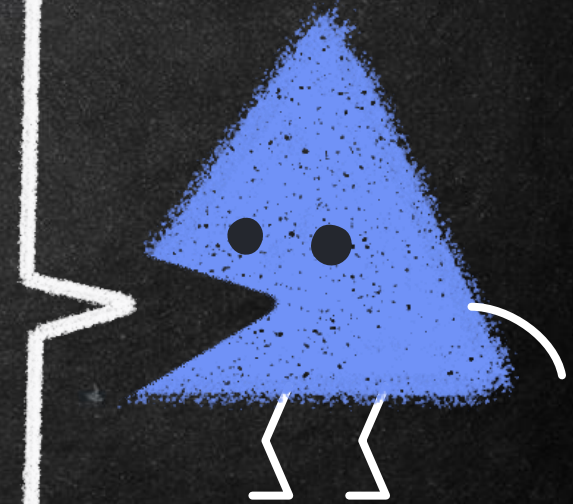
How Data is stored?

- computer's operating system only stores data as clusters.
- **Slack Space:** The slack space, is the difference between the space that is assigned and the space that is actually used.
- The hard drive is used to store your data and applications when they aren't being used.
- The sectors containing leftover data are commonly referred to as **slack space** or **cluster slack**.



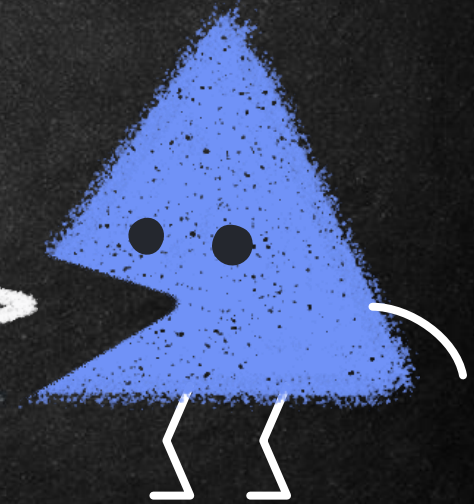
Slack Space using Example

- If we created a one-byte file, an NTFS file system would write the single byte and 511 null bytes to the cluster.
- If the cluster size were equal to one sector, it would be written in the same way. If the cluster size were 4 Kb (eight sectors), then the file system would write the first byte of the file, followed by 511 null bytes into the first sector.
- The remaining sectors in the cluster would not be written to (or overwritten if they previously contained data from a deleted file).



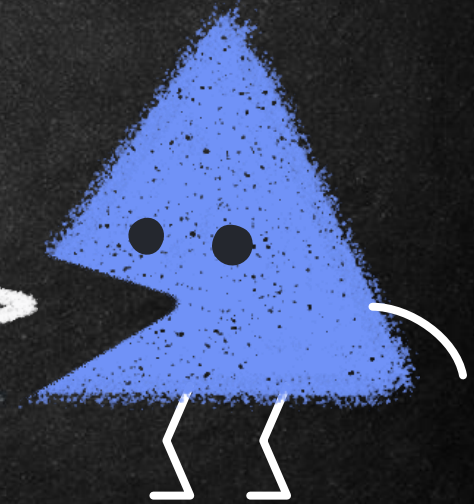
Data Carving

- Data carving is the act of searching for particular strings or bytes within a structure.
- A hex editor or other data-viewing tools can be used to carve for data.
- The analyst determines a string or binary pattern to search for, then initiates a search across a device or structure for that string or pattern.



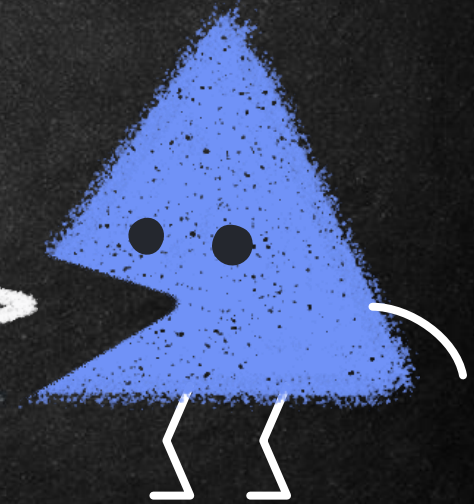
Data Carving

- The target can be of whatever scope is appropriate for the task, such as a file, slack space, unallocated space, a full volume, a memory image, or a swap file.
- The technique can be used to carve for full files – such as recovering deleted JPG image files; or for records – such as recovering portions of a deleted Windows event log.



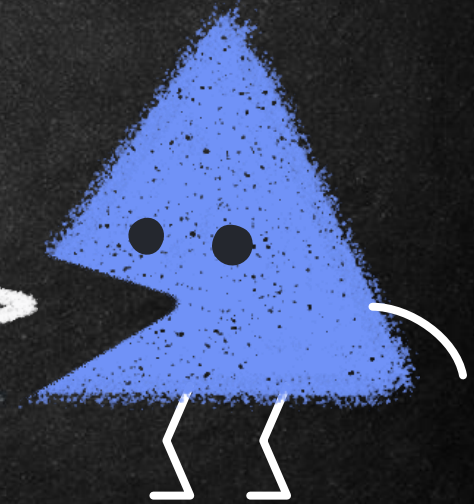
Data Carving

- Carving for records can be a very effective method of finding evidence when it's been deleted either intentionally or through normal system processes.



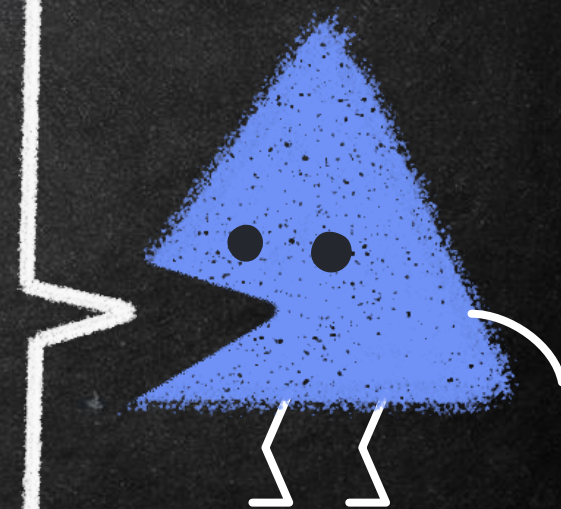
Data Carving : Files

- Full files can be carved by searching for particular file headers or signatures.
- File carving is typically limited to carving for contiguous – non fragmented – files in unallocated space or in other files.
- Techniques exist to carve data that is fragmented using fuzzy hashing



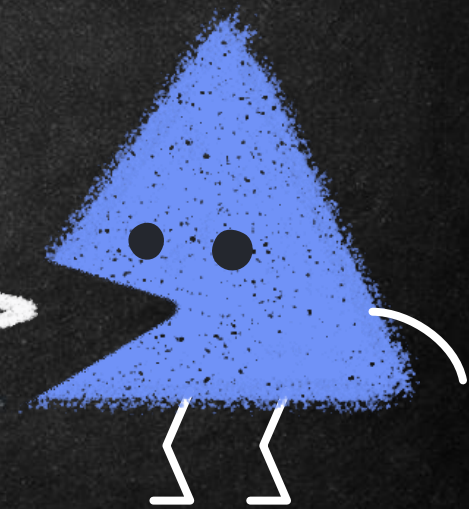
Data Carving : Records

- Carving for records involves searching for data where the full file may not be essential to recovering information.
- Like to carve for a weblog entry that contains full context in a single event: date and time, webpage visited, IP address accessing the page, browser ID, and so on.
- While a weblog is typically maintained in plain text, other data may be contained in binary blobs (such as Windows event logs) or a database format.



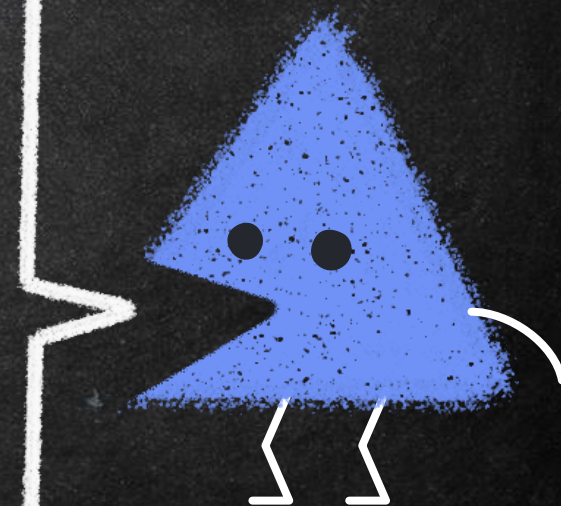
Data Carving: Example

- The forensic investigator received a CPU from law enforcement for examination.
- The system was seized because it was running a Windows System.
- When the system was analyzed, the system logs had been deleted and very little other logical evidence remained.
- The investigator carved for Windows Event Log records and successfully recovered over twenty two



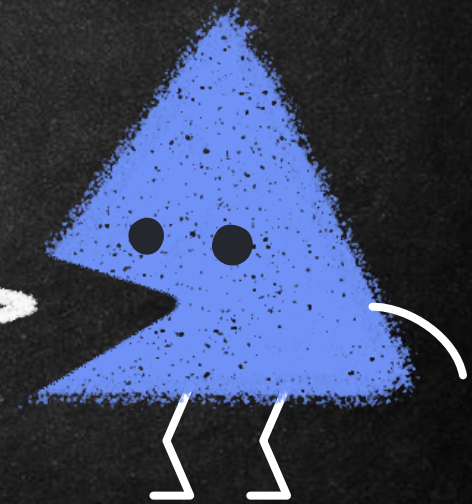
Data Carving: Example

- These entries were key in helping law enforcement determine the evidence for intrusion in the case because they contained information about all malware infections from the infected system checking into the system logs.
- Information retrieved included:
 - IP address of remote server,
 - system host name, and
 - date and time of the check-in etc.



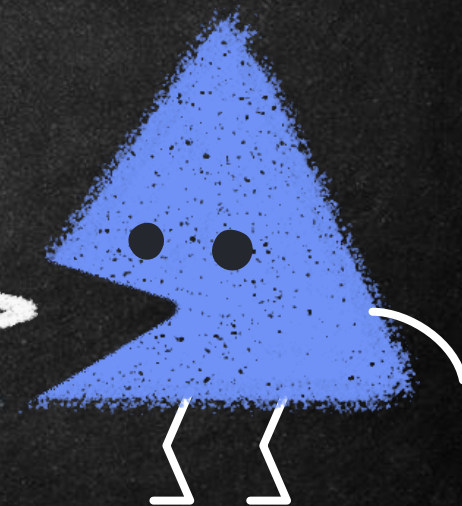
Data Acquisition

- Data acquisition is the act of collecting data relevant to an investigation.
- This data can be live data, including memory captures, or it can be collected in a forensically sound manner as an image or a forensic copy.
- The data acquisition should be conducted in a manner that creates the fewest changes to a system.



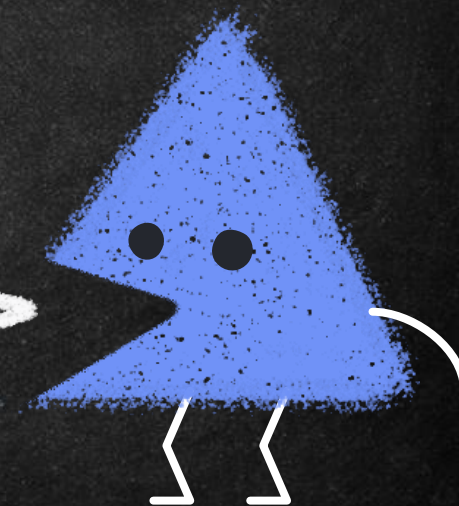
Data Acquisition

- The acquisition of a system that is powered off is the simplest example of collecting data in a forensically sound manner:
 - The analyst removes the system's hard drive.
 - The hard drive is connected to the read-only port of a firmware write block.
 - A target device to store an image or a copy is connected to the read-write port of the write block.



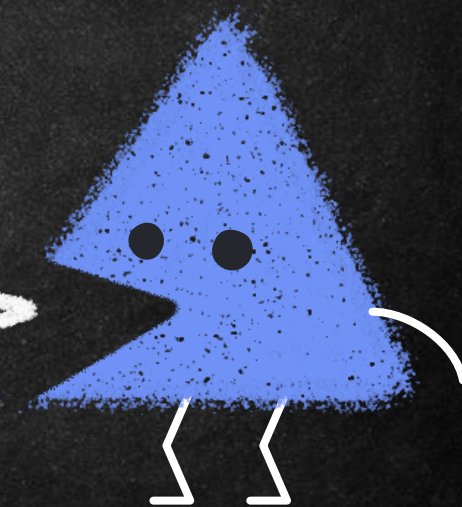
Data Acquisition

- The communications port of the write block is connected to a computer or a stand alone device, or is built in.
- The analyst then creates an image or a copy by writing to the target device.
- The original is maintained and documented in a chain-of-custody form.



Data Acquisition

- The image or copy can then be examined (or additional copies made) on the analysis platform.
- This is typically done as a read-only operation as well, which allows for repeatability of any examination or experiments without modifying the data.



References

- Digital Forensics by André Årnes
- Computer Systems: Digital Design, Fundamentals of Computer Architecture and Assembly Language by Ata Elahi
- Practical Forensic Imaging: Securing Digital Evidence with Linux Tools by Bruce Nikkel
- A Practical Guide to Digital Forensics Investigations 2nd Edition by Darren R. Hayes

